Marwadi University
Marwadi Chandarana Group

Syllabus for Diploma Engineering

Electrical Engineering

Semester – VI

Subject Name: Energy Conservation Strategies and Auditing

Subject Code: 09EE0610

Diploma Branches in which this subject is offered: Electrical Engineering

Objective: The use of energy is increasing day by day. One way to cope up with the increase in energy requirement is to increase the generation of energy which requirements more investment and the other way is to conserve the energy because energy saved is energy generated. Energy conservation means decreases in energy uses but not compromising with the quality or quantity of energy generation. Must require theoretical and practical knowledge about the concept of energy conservation, energy management, and different approaches of energy conservation in industries, and energy audit and its measuring instruments in commercial and industrial sector will be clear fundamental by this course.

Credits Earned: 3 Credits

Course Outcomes: After completion of this course, student will be able

1. Understand energy problem, conservation concepts, and scope for energy efficiency in different areas.
2. Understand how to apply energy-saving techniques in lighting, heating, cooling, motors, and other electrical systems.
3. Analyze tariffs, calculate costs of energy conservation project.
4. Understand energy conservation practices in power generation, transmission, and distribution systems.
5. Identify process of energy audit and measures for energy conservation.

Pre-requisite of course: Basic knowledge of D.C. Circuits, A.C. Circuits, Electrical Machine, Power system and Electrical measurement

Teaching and Examination Scheme

Teaching Scheme (Hours)			Credits	Theory Marks			Tutorial/ Practical Marks		Total Marks
Theory	Tutorial	Practical		ESE	IA	CSE	Viva	Term work	
2	0	2	3	50	30	20	25	25	150

Contents:

Unit	Topics	Contact hours	Weightage (%)
1	**Energy** • Introduction • Energy, Energy problem in India • Review of various energy sources • Need and concept of energy conservation • Energy conservation principle – maximum energy efficiency, maximum cost effectiveness • Power utilization in India, Scope of energy conservation indifferent area • Bureau of energy efficiency • Energy management concept and objectives • Principles of energy management, different approach of energy management, elements of energy management	**03**	**11**
2	**Energy Conservation** • Introduction • Conservation of electric power • Lighting energy: methods/techniques of efficient lighting. • Heating: methods/techniques of energy saving in Furnaces, Ovens and Boilers. Energy conservation by area sealing, use of thermal insulation • Cooling: methods/techniques of energy saving in ventilating systems and air conditioners • Energy conservation in air compressor, pumps, fan & blower, welding equipment • Motive power • Use of energy efficient motors and efficient use of energy in motors with the help of voltage reducers, automatic star/ delta converters. Reduction of losses in MCC, effect of harmonics on motor, energy conservation by VSD and PAM • Power factor improvement devices and soft starters/variable frequency drives. • Amorphous core transformers • Energy conservation act	**08**	**29**
3	**Tariff and Energy Conservation in Industries** • Introduction • Need of energy efficient devices • Energy cost and recent MSEB tariffs • Application of tariff system to reduce energy bill	**04**	**14**

	• Energy conservation by improving load factor and power factor. • Calculation and costing of energy conservation project • Different cost of a project, interest cost, Depreciation, Risk analysis, modes of economic analysis, Case study		
4	**Energy Conservation in Generation, Transmission and Distribution Systems** • Introduction • Energy conservation in power plant, main elements and equipment of power plants • Improvement in existing power plant • Combined cycle power plant • Cogeneration plants - types and advantages. • Small hydro power plants • Demand side management • Reactive power compensation • Promote renewable energy • System voltage optimization and phase current balancing • Losses in transmission and distribution system and its minimization	**05**	**17**
5	**Energy Audit** • Introduction • Energy audit and its procedure, benefits and elements of energy audit • Energy Flow diagram and its importance • Types and methodology of energy audit • Measurements in energy audit and various measuring instruments • Questionnaires for the energy audit • Internal energy audit checklist • Equipment used for energy conservation • Calculation of payback period for energy conservation equipment. • Procedure to carryout energy audit of induction motor and transformer • IE rules and regulations for energy audit • Electricity act 2003(Numerical)	**08**	**29**

Suggested Theory distribution:

The suggested theory distribution as per Bloom's taxonomy is as per follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process

Distribution of Theory for course delivery and evaluation					
Remember	Understand	Apply	Analyze	Evaluate	Create
40%	40%	10%	10%	0%	0%

Suggested List of Experiments:

Sr. No.	Unit No.	Name of Topics	Contact Hours
1	1	State different energy management systems prevailing in a particular organization and industry.	2
2	1	To prepare an energy management program in a given organization.	4
3	2	Calculation of energy savings and cost analysis for energy conservation using lighting devices.	4
4	2	Calculation of energy savings and cost analysis for energy conservation for cooling devices.	2
5	3	Collect the standard tariff rates and suggest suitable tariff for given industry/lab/institute/commercial establishment.	2
6	3	Find depreciation cost of any one given energy conservation project/equipment.	2
7	4	State the different energy conservation methods useful in power generation, transmission and distribution.	2
8	5	To do assessment of environmental impact of energy utilization.	4
9	6	State importance of energy flow diagram and draw the energy flow diagram for an industry/organization division.	2
10	6	Find out the payback period for a given energy conservation equipment.	2
11	6	Prepare a technical report on energy conservation act 2003.	2

Instructional Method:

a. The course delivery method will depend upon the requirement of content and need of students. The teacher in addition to conventional teaching method by black board, may also use any of tools such as demonstration, role play, Quiz, brainstorming, MOOCs etc.

b. The internal evaluation will be done on the basis of continuous evaluation of students in the laboratory and class-room.

c. Practical examination will be conducted at the end of semester for evaluation of performance of students in laboratory.

d. Students will use supplementary resources such as online videos, NPTEL videos, e-courses.

References:

1. C.L.Wadhawa, "*Generation Distribution and Utilization of Electrical Energy*", New Age. 2012.
2. Steven R. Patrick, Dale R. Patric and Stephen W. Fardo, "*Energy conservation Guide book*", Fairmont Press. 2011.
3. S Sivaganaraju, "*Electric Energy Generation, Utilization and Conservation*", Pearson, New Delhi, 2012.
4. H. Partab, "*Art and Science of utilization of Electrical Energy*", Dhanapat Rai and Sons, New Delhi. 2011.
5. Wayne C. Turner, "*Energy Management Handbook*", Fairmont Press. 2012.
6. Albert Thumann, Terry Niehus and William J. Younger, "*Handbook of Energy Audits*", Fairmont Press. 2013.
7. Y.P.Abbi, "*Handbook on Energy Audit and Environment Management*", Teri Press. 2014.

Supplementary Resources:

1. https://nptel.ac.in/course.php
2. www.bee. com
3. www.beeindia.in
4. https://beeindia.gov.in/content/rules-and-regulations
5. http://www.cercind.gov.in/Act-with-amendment.pdf
6. https://powermin.nic.in/en/content/electricity-act-2003